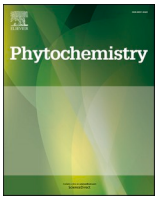
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Identification of seven undescribed cucurbitacins in *Cucumis sativus* (cucumber) and their cytotoxic activity

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ABSTRACT

Cucurbitacin C-type cucurbitacins that are only identified in *Cucumis sativus* (cucumber) are, in part, responsible for the health benefits and bitter flavor. Nevertheless, detailed information about those functional ingredients in cucumber is scarce. In this study, ten cucurbitacin C analogues including seven undescribed ones have been isolated from the bitter leaves of cucumber, in which six compounds showed growth inhibition capabilities against tumor cell lines HepG2, A549, DU145 and HCT116. Intriguingly, cucurbitacin C6 and C7 exhibited a significant inhibition effect compared to the positive control taxol (IC_{50} = 1.86 ± 0.17 μM) on HepG2 cell line with IC_{50} values of 10.06 ± 0.34 μM and 4.16 ± 0.42 μM, respectively. The mechanism of cucurbitacin-induced apoptosis is likely down-regulating the expression of caspase-related proteins. This work enlarges the knowledge of the cucurbitacins in cucumber and highlights the importance of cucumber as a source of specialized metabolites in the food and medicinal industries.

1. Introduction

The cucumber (*Cucumis sativus* L, Cucurbitaceae), a popular fruit and vegetable, has made a tremendous contribution to providing a sufficient and high-quality food supply. For decades, identification and bioactivity evaluation on most cucurbitacin categories have been well documented except for cucurbitacin C (CuC) and its analogues, which are the primary bitter principle in cucumber, and solely discovered in this crop (Enslin et al., 1960; Qing et al., 2014; Shang et al., 2014). Compared to the structures of well-studied cucurbitacins, such as cucurbitacin B, cucurbitacin E and hemslecin A (also known as 25-acetoxy-23,24-dihydrocucurbitacin F), CuC shares most of the similarities, yet, exhibits discriminable structural diversities at its C-2, C-3 and C-19 positions (Fig. 1). Recently, the potential anti-cancer activity of CuC to human cancers has been explored both *in vitro* and *in vivo* and revealed that PI3K-Akt signaling pathway might be the major pathways involved in CuC-induced cell growth arrest and apoptosis (Wu et al., 2019), whereas, numerous of its potent analogues have not been found thus far. Hence, it is essential to investigate the bioactive molecular entities of CuC analogues systemically from the cucumber plants.

Cucurbitacins are characterized by a tetracyclic triterpene skeleton that is further oxidized and glycosylated by cytochrome P450s (CYP450s) and glycosyltransferases, respectively, to form the structurally diverse triterpenoids (Chen et al., 2005; Zhao et al., 2017). Based on molecular structural differences, twelve categories of cucurbitacins have been discovered so far, most of which were identified from the Cucurbitaceae family, such as *Cucumis*, *Trichosanthes*, *Momordica*, *Hemsleya*, *Siraitia* (Cai et al., 2015; Chen et al., 2005). These extremely bitter compounds (except for several mogrosides) exhibit a broad range of pharmacological properties, including anti-inflammatory, antipyretic,

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and anti-cancer activities (Lee et al., 2010; Wu et al., 2019). In particular, their potential roles as specific inhibitors of the JAK-STAT pathway have been extensively studied (Chan et al., 2010; Deng et al., 2016; Kong et al., 2014; Sun et al., 2005; Thoennissen et al., 2009). Moreover, specific cucurbitacins prepared from *C. melon* (cucurbitacin B) and *H. amabilis* (hemslecin A) have been clinically used as prescription drugs to treat hepatic tumor and infection disease, respectively, in China (Lu et al., 2018). However, the medicinal use of cucurbitacins has been hampered by their structural complexity and low abundance in natural sources, which limit bulk chemical synthesis or planta extraction (Jung et al., 2010; Ren et al., 2016). In addition, non-specific cytotoxicity of cucurbitacins also leads to their limited usage and being pursued under strict medical control (Chen et al., 2005).

After a long period of domestication, fruits in most commercial cucumber germ lines are non-bitter, however, the bitterness was highly conserved in the leaves. Here, we report the isolation and structural identification of seven undescribed CuC type cucurbitacins (**1**–**5** and **9**–**10**) along with three known ones from the bitter leaves of cucumber. Interestingly, two undescribed compounds (**9** and **10**) exhibit significant antitumor activities on four cancer cell lines, including HepG2, A549, DU145 and HCT116. These discoveries would enrich the candidates for anticancer drug development and provide new insight to make full use of cucumber resources.

2. Results and discussion

2.1. Isolation and structural identification of cucurbitacins from cucumber leaves

Compared to other categories, the CuC type cucurbitacins are rare and only identified in cucumber (Chen et al., 2005). Thus, to enlarge the knowledge of the cucurbitacins in cucumber, an inbred cucumber line (9930) with extremely bitter leaves (indicating high content of cucurbitacins) was used in this study. Ten compounds (Fig. 2) were isolated and identified to be C-19 hydroxylated cucurbitacins through a stepwise chromatographic separation and extensive spectroscopic studies. Three compounds including deacetylcucurbitacin C (**6**), 23,24-dihydrocucurbitacin C (**7**) and cucurbitacin C (**8**), which were considered the major cucurbitacins in cucumber, were identified as previously reported components (Qing et al., 2014; Shang et al., 2014). Six undescribed compounds including cucurbitacin C2 (**2**), cucurbitacin C3 (**3**), cucurbitacin C4 (**4**), cucurbitacin C5 (**5**), cucurbitacin C6 (**9**) and cucurbitacin C7 (**10**), were speculated to be the post-modified derivatives of CuC or deacetylcucurbitacin C. Cucurbitacin C1 (**1**) is structurally considered to be a by-pass product during CuC biosynthesis.

Compound **1** was obtained as a white amorphous powder with $[\alpha]_D^{20}$ +99.2 (*c* 0.10, MeOH). The molecular formula was established as $C_{30}H_{46}O_5$ based on the quasi-molecular ion peak of the high-resolution electrospray ionization mass spectrometry (HRESIMS) at *m/z* 487.3409 $[M+H]^+$ (Calculated for 487.3418, Fig. S1A), along with the ^{13}C nuclear magnetic resonance (NMR) spectrum (Fig. S2B). The ratio of carbons to hydrogen in the molecule indicated eight double bond equivalents. The IR spectrum (Fig. S1B) of compound **1** showed strong absorption brands at 3416 and 1697 cm^{-1} suggesting of hydroxy and carbonyl functionalities. The ^1H and ^{13}C NMR data (Tables 1 and 2) assigned by HSQC (Fig. S3A) and ^1H–^1H COSY experiments (Fig. S4A) suggested compound **1** was a cucurbitane-type triterpene (Qing et al., 2014; Shang et al., 2014). The ^1H NMR data showed the presence of seven tertiary methyls [δ_H 0.90, 1.03, 1.16, 1.25, 1.29, 1.64, 1.71, (3H each, s)], two olefinic protons (δ_H 5.74, d, J = 6.0 Hz; 5.90, d, J = 8.0 Hz). As there was one carbonyl signal (δ_C 213.7), two pairs of olefinic carbons (δ_C 141.6/119.9, 135.4/125.1) and five resonances (δ_C 76.6, 73.8, 71.4, 70.2, 61.4) for carbons bearing singly bonded oxygen in the ^{13}C NMR spectra, the combined data suggested an additional ring formed by cyclization of the side chain through an ether linkage. The 1D NMR spectrum of **1** showed close resemblances to that reported for 16*α*, 23*α*-epoxy-2*β*,3*β*,20*β*-trihydroxy-23*α*-cucurbit-5,24-dien-11-one (Schenkel et al., 1992), the only difference that the 19-methyl was hydroxylated in **1** whereas the hydroxylation was taken place at the C-2 position in the latter. These variances in **1** could be confirmed by ^1H–^1H COSY correlations from H-1 (δ_H 0.95) to H-2 (δ_H 1.39 and 1.77), from H-2 to H-3 (δ_H 3.20), and by HMBC correlations (Fig. 3 and Fig. S3B) from H-19 (δ_H 3.18 and 4.31) to C-11 (δ_C 213.7), C-9 (δ_C 54.9) and C-10 (δ_C 33.6). The ether linkage was confirmed between C-16 and C-23 by critical HMBC correlation from H-23 (δ_H 4.68) to C-16 (δ_C 70.2). The cross-peaks at CH_3-30/H-17 (δ_H 1.97) and H-17/CH_3-21 (δ_H 1.29) observed in NOESY experiment (Fig. 3 and Fig. S4B) revealed *α*-orientation of CH_3-30, H-17 and CH_3-21 (20-OH was *β*-orientation), which was in good accordance with that of the reported datiscoside (cucurbitacin D glycoside) (Kupchan et al., 1972). The *α* and *S*-configuration of H-23 and C-23, respectively, were deduced by the observation of the NOESY correlations of H-23/H-17, and H-16 (δ_H 4.57)/H-24 (δ_H 5.90), which was in agreement with the NOE behavior of the E ring in 16*α*, 23*α*-epoxy-2*β*,3*β*,20*β*-trihydroxy-23*α*-cucurbit-5,24-dien-11-one and contrasted to those of the isomer 16*α*,23*α*-epoxy-2*β*,3*β*,20*β*-trihydroxy-23*β*-cucurbit-5,24-dien-11-one (Schenkel et al., 1992). Thus, the compound **1** was elucidated as 23(*S*)-3*β*,19,20*β*-trihydroxy-16*α*, 23*α*-epoxy-cucurbit-5,24-dien-11-one, and was defined the name cucurbitacin C1 (Fig. 2).

Compound **2** was obtained as a white amorphous powder with $[\alpha]_D^{20}$ +70.0 (*c* 0.10, MeOH). Compound **2** gave a quasimolecular ion at *m/z* 537.3420 $[M+H]^+$ (Calculated for 537.3427, Fig. S5A) in the HRESIMS, corresponding to the formula $C_{30}H_{48}O_8$. The IR spectrum of **2** (Fig. S5B) showed strong absorption brands at 3434 and 1689 cm^{-1} suggesting

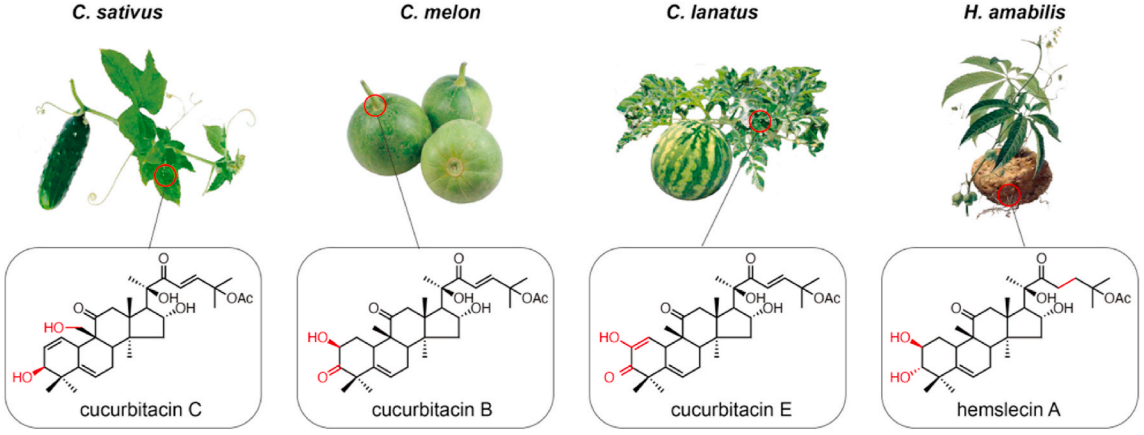


Fig. 1. Typical cucurbitacins isolated from the planta of Cucurbitaceae.

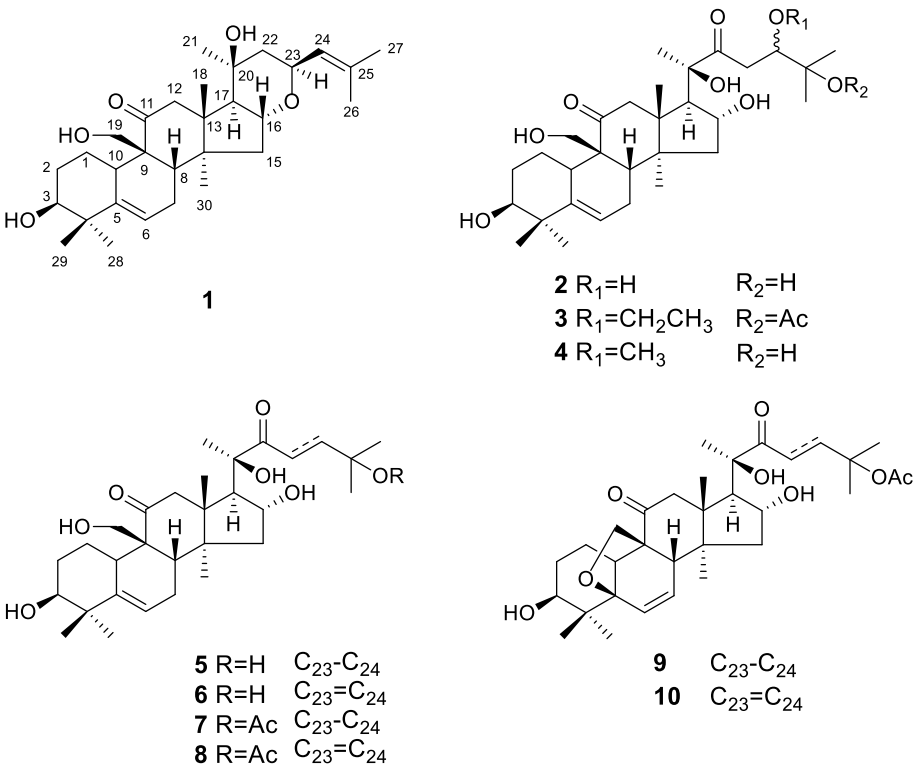


Fig. 2. Cucurbitacins isolated from the leaves of cucumber.

Table 1
^{1}H NMR data of compounds 1–5, 9 and 10 (500 MHz). Compounds **4**–**5** were measured in C_5D_5N, the others were in $CDCl_3$. δ in ppm, J in Hz.

No.	1	2	3	4	5	9	10
1	1.64, m;	1.61, m;	1.66, m;	2.07, m[a];	2.10, m[a];	1.36, m;	1.34, m;
	0.95, m	0.81, m	0.94, m	1.16, m	1.15, m	1.03, m	1.05, m
2	1.77, m;	1.67, m;	1.79, m;	2.04, m[b];	2.05, m;	1.67, m;	1.69, m;
	1.39, m	1.33, m	1.38, m	1.85, m	1.84, m	1.48, m	1.46, m
3	3.20, m	3.09, m	3.21, dd (11.3, 4.0)	3.50, m	3.51, dd (11.0, 4.4)	3.76, dd (10.5, 2.8)	3.77, dd, (10.5, 3.0)
6	5.74, d (6.0)	5.59, d (4.0)	5.74, d (5.5)	5.83, d (5.5)	5.83, d (5.5)	6.15, dd (9.8, 2.0)	6.15, dd (9.8, 1.5)
7	2.40, m;	2.31, m;	2.39, m;	2.64, m;	2.65, m;	5.63, dd (9.8, 3.5)	5.65, dd (9.8, 3.0)
	1.91, m[a]	1.79, m	1.92, m [a]	2.04, m[b]	2.09, m[a]		
8	2.67, d (7.5)	2.68, d (7.2)	2.65, d (7.5)	3.31, m[c]	3.27, m	2.48, br s	2.48, br s
10	2.27, m	2.19, m	2.32, m	2.62, m	2.62, m	2.41, m	2.41, m[a]
12	3.02, d (14.5);	3.05, d (14.5);	3.20, d (14.2);	3.29, d (14.5)[c];	3.30, d (14.7);	2.92, d (14.5);	2.91, d (14.5);
	2.42, d (14.5)	2.48, d (14.5)	2.61, d (14.2)	2.90, d (14.5)	2.90, d (14.7)	2.71, d (14.5)	2.70, d (14.5)
15	1.90, m[a];	1.81, m;	1.90, m[a];	2.08, m[a];	2.11, m;	1.93, d (13.5);	1.97, d (14.5);
	1.42, m	1.37, m	1.43, m	1.83, m	1.80, m	1.49, d (13.5)	1.52, d (14.5)
16	4.57, m	4.34, m	4.34, m	4.98, m	4.95, m	4.35, m	4.41, m
17	1.97, d (9.5)	2.60, d (7.0)	2.43, d (7.0)	3.01, d (7.0)	3.04, d (7.0)	2.43, d (7.0)	2.42, d (7.0) [a]
18	1.03, s	0.99, s	1.03, s	1.62, s	1.63, s	1.39, s	1.39, s
19	4.31, d (11.0);	4.29, d (10.5);	4.31, d (10.8);	4.87, d (10.0)	4.87, d (10.5);	4.64, d (8.5);	4.64, d (8.5);
	3.18, d (11.0)	3.07, d (10.5)	3.18, d (10.8)	3.59, d (10.0)	3.56, d (10.5)	3.37, d (8.5)	3.38, d (8.5)
21	1.29, s	1.30, s	1.41, s	1.69, s	1.65, s	1.39, s	1.42, s
22	1.87, dd (14.5, 7.0);						
	1.47, d (14.5)						
23	4.68, dd (8.0, 7.0)	2.95, dd (16.5, 9.0);	2.87, dd (17.2, 1.5);	3.85, dd (17.0, 2.5);	3.54, m;	2.80, m;	6.43, d (15.0)
		2.51, d (16.5, 1.5)	2.76, dd (17.2, 9.5)	3.35, dd (17.0, 8.5)	3.33, m	2.48, m	
24	5.90, d (8.0)	3.94, dd (9.0, 1.5)	4.21, dd (9.5, 1.5)	4.11, dd (8.5, 2.5)	2.28, m; 2.20, m	2.03, m	7.03, d (15.0)
26	1.64, s	1.11, s	1.52, s	1.46, s	1.40, s	1.42, s	1.54, s
27	1.71, s	1.12, s	1.43, s	1.44, s	1.40, s	1.45, s	1.56, s
28	1.16, s	1.07, s	1.17, s	1.48, s	1.47, s	1.11, s	1.12, s
29	0.90, s	0.86, s	0.95, s	1.32, s	1.32, s	0.87, s	0.87, s
30	1.25, s	1.26, s	1.33, s	1.65, s	1.65, s	0.94, s	0.95, s
CH$_3$CO-			1.95, s			1.96, s	2.01, s
CH$_3$CH$_2$O-			1.08, t (7.0)				
CH$_3$**CH$_2$**O-			3.62, m				
CH$_3$O-				3.56, s			

a, *b*, *c*, these signals were overlapped.

Table 2
^{13}C NMR data of compounds 1–5, 9 and 10 (125 MHz). Compounds **4**–**5** were measured in C_5D_5N, the others were in $CDCl_3$. δ in ppm.

No.	1	2	3	4	5	9	10
1	24.6	24.2	24.6	25.6	25.8	23.1	23.1
2	30.8	30.8	30.8	32.3	32.3	28.9	29.0
3	76.6	75.6	76.5	76.4	76.4	73.0	73.1
4	42.4	42.3	42.4	43.4	43.4	39.6	39.6
5	141.6	141.9	141.6	143.4	143.4	86.8	86.8
6	119.8	118.9	119.7	119.8	119.8	133.3	133.3
7	23.8	23.4	23.7	24.5	24.5	129.0	129.1
8	34.9	33.0	33.6	33.9	33.9	51.9	52.0
9	54.9	53.8	54.4	54.6	54.6	58.8	58.8
10	33.6	35.1	35.1	36.2	36.2	40.4	40.5
11	213.7	212.5	213.5	212.6	212.7	208.8	208.8
12	48.4	48.6	48.8	49.9	49.9	50.7	50.7
13	48.5	50.5	51.2	51.5	51.4	50.1	50.2
14	48.1	47.9	48.1	48.8	48.8	47.5	47.3
15	41.3	45.4	45.8	47.2	47.2	44.4	44.3
16	70.2	70.2	71.4	70.9	70.9	71.4	71.8
17	55.5	55.7	57.7	59.1	59.5	58.1	58.5
18	18.6	18.4	18.8	19.9	20.2	20.9	21.1
19	61.4	60.0	61.3	60.8	60.8	75.2	75.3
20	73.8	79.3	79.2	80.7	80.6	78.9	78.2
21	29.8	23.3	24.3	25.7	25.3	22.5	24.1
22	46.4	214.1	212.9	215.3	216.6	214.0	202.6
23	71.4	39.6	38.6	40.0	33.1	30.6	120.5
24	125.0	74.5	79.9	85.8	38.9	34.8	152.1
25	135.3	71.5	83.9	72.7	69.4	81.4	79.4
26	18.1	25.0	21.9	25.5	30.4	26.0	26.0
27	26.2	25.4	22.7	25.2	30.4	26.6	26.5
28	24.5	24.4	24.5	27.9	25.6	19.5	19.4
29	20.3	20.2	20.4	21.1	21.4	16.6	16.6
30	20.3	19.3	19.6	20.1	19.8	20.0	20.0
CH_3CO-			22.8			22.5	22.1
CH_3CO-			170.4			170.5	170.4
CH_3 CH_2O-			16.0				
CH_3**CH_2**O-			68.9				
CH_3O-				60.5			

hydroxy and carbonyl functionalities. The NMR spectra (Tables 1 and 2, Fig. S6) assigned by HSQC (Fig. S7A) and ^{1}H–^{1}H COSY experiments (Fig. S8A) showed the presence of seven tertiary methyls [δ_H 0.86, 0.99, 1.07, 1.11, 1.12, 1.26, 1.30 (3H each, s)], one olefinic proton (δ_H 5.49, d, J = 4.0 Hz); two carbonyl carbon resonances at δ_C 212.5 and 214.1; a pair of olefinic carbon at δ_C 141.9/118.9 and six oxygenated methines at δ_C 79.3, 75.6, 74.5, 71.5, 70.2, 60.0. Comparing of the NMR data of **2** with those of deacetylcucurbitacin C (Shang et al., 2014) indicated that compound **2** was a derivative of deacetylcucurbitacin C. The primary variance between the two compounds was that a pair of olefinic carbon signals denoting the double bond in the side chain was absent from **2**, indicating the saturation of this double bond in **2**. Additionally, one more oxygenated carbon signal (δ_C 75.4) was observed in **2**, which was then placed on C-24 position by correlations from H-23a (δ_H 2.95) to H-24 (δ_H 3.94) in ^{1}H–^{1}H COSY experiment, and from CH_3-26 (δ_H 1.11) and CH_3-27 (δ_H 1.12) to C-24 (δ_C 74.5) in HMBC experiment (Fig. S7B). To determine the stereochemistry of C-24, computational calculation using DFT at the B3LYP-D3(BJ)/6-31G (d) level was applied to calculate the chemical shift of H-23 and the dihedral angle of H-23/H-24 in enantiomers 24*R*-**2** and 24*S*-**2**. As a result, the absolute values of the dihedral angle of H-23a (δ_H 3.16)/H-24 and H-23 b (δ_H 2.49)/H-24 in 24*R*-**2** were calculated to be 78.0° and 165.1°, respectively; whereas in 24*S*-**2**, the values of H-23a (δ_H 3.04)/H-24 and H-23 b (δ_H 2.62)/H-24 were 175.6° and 65.7°, respectively (Fig. S9). Based on this information, the proton-proton coupling constant of H-23/H-24 in the corresponding enantiomer was calculated by Karplus-equation (Karplus, 1963), and the computational J value in 24*S*-**2** (Fig. S9) well matched with the experimental value of H-23a (δ_H 2.95, J = 9.0 Hz) and H-23 b (δ_H 2.51, J = 1.5 Hz), thus, the absolute configuration of C-24 in **2** was deduced to be *S*. Consequently, compound **2** was elucidated as 24(*S*)-3β,16α,19,20β,24,25-hexahydroxycucurbit-5-en-11,22-dione (cucurbitacin C2).

Compound **3** was a white amorphous powder with $[\alpha]_D^{20}$ +54.7 (*c* 0.10, MeOH). The molecular formula of **3** was established as $C_{34}H_{54}O_9$ based on the quasi-molecular ion peak of HRESIMS (positive mode) at *m/z* 607.3835 $[M+H]^+$ (Calculated for 607.3841, Fig. S10A), along with the ^{13}C NMR data. The IR spectrum (Fig. S10B) of compound **3** showed strong absorption brands at 3393, 1726 and 1700 cm^{-1} suggesting hydroxy, acyl carbonyl and carbonyl functionalities. The NMR data (Tables 1 and 2, Fig. S11) assigned by the HSQC experiment (Fig. S12A) showed the presence of eight tertiary methyls [δ_H 0.95, 1.03, 1.17, 1.33,

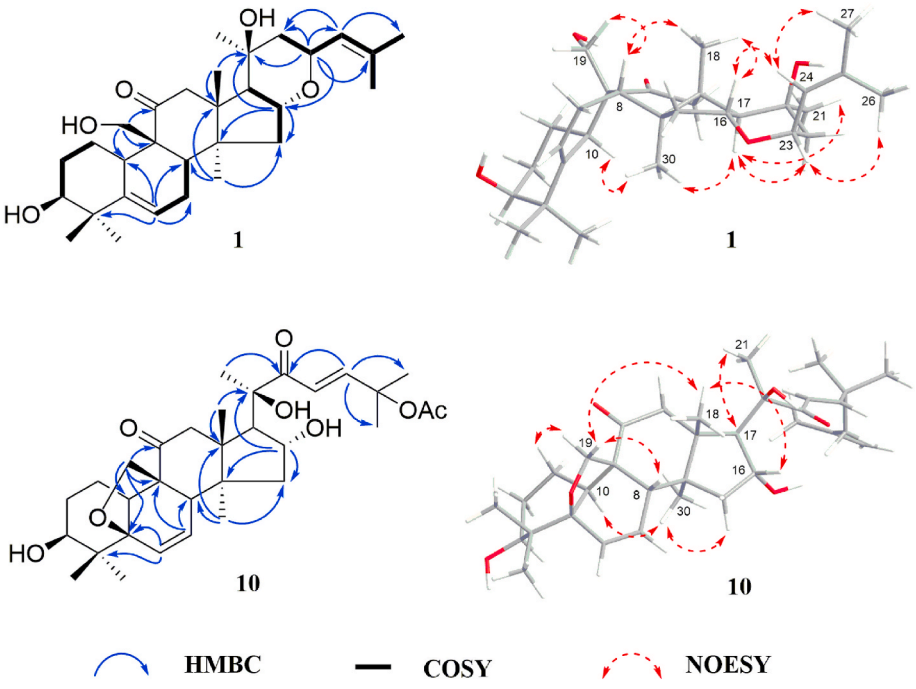


Fig. 3. Key HMBC,^{1}H–^{1}H COSY, NOESY correlations and energy-minimized 3D structures of **1** and **10**.

1.41, 1.43, 1.52, 1.95 (3H each, s)], one primary methyl [δ_H 1.08 (3H, t, J = 7.0 Hz)], one olefinic proton (δ_H 5.74, d, J = 5.5 Hz); three carbonyl carbon resonances at δ_C 170.4, 212.9, 213.7; a pair of olefinic carbon at δ_C 141.6/119.7; seven oxygenated methines at δ_C 83.9, 79.9, 79.2, 76.5, 71.4, 68.9, 61.3. These 1D NMR data suggested **3** was a derivative of 23,24-dihydrocucurbitacin C (Qing et al., 2014), as the resonance signals between the two compounds are highly similar. The only difference was the presence of an ethoxy moiety in the structure of **3**, which was substantiated by one primary methyl ^{1}H resonance [δ_H 1.08 (3H, t, J = 7.0 Hz)] and two hydroxyl-carbon ^{13}C resonances (δ_C 68.9 and 79.9). By comparison of the ^{13}C NMR data between **3** and 23,24-dihydrocucurbitacin C, the ethoxy functionality was deduced to attaching on C-24 position based on the ^{13}C resonances of C-23, C-24 and C-25 in **3** moving to the low-field by Δ+8.0, +44.9, +2.7 ppm, respectively. This conclusion was further confirmed by the HMBC experiment (Fig. S12B), which showed the cross-peaks from H-24 (δ_H 4.21) to C-23 (δ_C 38.6), C-25 (δ_C 83.9), C-22 (δ_C 212.9) and the ethoxy C-1′ (δ_C 68.9); and from the ethoxy H-1′ (δ_H 3.62) to C-24 (δ_C 79.9). The proton coupling constant pattern of H-23 (2.87, dd, J = 17.2, 1.5 Hz; 2.76, d, J = 17.2, 9.5 Hz) in **3** was in contrast with those in compound **2**, suggesting the *R* configuration of C-24. Based on the above evidence, compound **3** was elucidated as 24(*R*)-ethoxy-25-acetyloxy-3*β*,16*α*,19,20*β*-tetrahydroxycucurbit-5-en-11, 22-dione, and was defined the name cucurbitacin C3.

Compound **4** was a white amorphous powder with $[\alpha]_D^{20}$ +112.0 (*c* 0.10, MeOH). The quasi-molecular ion peak of HRESIMS (positive mode) at *m/z* 551.3574 $[M+H]^+$ and ^{13}C NMR data of compound **4** were consistent with the molecular formula $C_{31}H_{50}O_8$ (Calculated for 551.3578, Fig. S14A). The IR spectrum (Fig. S14B) of compound **4** showed strong absorption brands at 3435, and 1694 cm^{-1} suggesting hydroxy and carbonyl functionalities. The NMR data (Tables 1 and 2, Fig. S15) assigned by HSQC experiment (Fig. S16A) showed the presence of seven tertiary methyls [δ_H 1.32, 1.44, 1.46, 1.48, 1.62, 1.65, 1.69 (3H each, s)], one methoxy [δ_H 3.56 (3H, s)], one olefinic proton [δ_H 5.83 (1H, d, J = 5.5 Hz)]; two carbonyl carbon resonances appeared at δ_C 212.6, 215.3; one pair of olefinic carbon at δ_C 143.4/119.8; five oxygenated methines at δ_C 85.8, 80.7, 76.4, 72.4, 60.8. These 1D NMR data suggested **4** was an analogue of compound **2**. A careful comparison of the ^{13}C NMR data between **4** and **2** revealed the only difference that the 24-OH was etherified by a methyl group in **4**. This variation could also be confirmed by the HMBC experiment (Fig. S16B), which showed the correlations from H-24 (δ_H 4.11) to the methoxy carbon (δ_C 60.5); and from the methoxy proton (δ_H 3.56) to C-24 (δ_C 85.8). Otherwise, the absolute configuration of C-24 in **4** was concluded to be *R* by the proton coupling constant pattern of H-23 (3.85, dd, J = 17.0, 2.5 Hz; 3.35, d, J = 17.0, 8.5 Hz) in **4**, which conformed with those in compound **3** and contrary to that in **2**. Thus, compound **4** was elucidated as 24(*R*)-methoxy-3*β*,16*α*,19,20*β*,25-pentahydroxycucurbit-5-en-11,22-dione, and was defined the name cucurbitacin C4.

Compound **5** was obtained as a white amorphous powder with $[\alpha]_D^{20}$ +85.9 (*c* 0.10, MeOH). The HRESIMS (positive mode) quasi-molecular ion peak at *m/z* 521.3471 $[M+H]^+$ and ^{13}C NMR data of compound **5** were consistent with the molecular formula $C_{30}H_{48}O_7$ (Calculated for 521.3473, Fig. S18A). The IR spectrum (Fig. S18B) of **5** showed strong absorption brands at 3399 and 1697 cm^{-1} suggesting hydroxy and carbonyl groups. The 1D NMR spectra (Tables 1 and 2, Fig. S19) assigned by HSQC experiment (Fig. S20A) showed the presence of seven tertiary methyls [δ_H 1.32, 1.40, 1.40, 1.47, 1.63, 1.65, 1.65 (3H each, s)], one olefinic proton [δ_H 5.83 (1H, d, J = 5.5 Hz)]; two carbonyl carbon resonances appeared at δ_C 212.7, 216.6; one pair of olefinic carbon at δ_C 143.4/119.8; five oxygenated methines at δ_C 80.6, 76.4, 70.9, 69.4, 60.8. These NMR data suggested that the structure of compound **5** was highly resemblance with deacetylcucurbitacin C (deacetyl CuC) (Shang et al., 2014), the only difference that the $\Delta^{23,24}$ double bond in the side chain of deacetyl CuC was saturated in **5**. Thus, compound **5** was elucidated as 3*β*,16*α*,19,20*β*,25-pentahydroxycucurbit-5-en-11, 22-dione (cucurbitacin C5).

Compounds **9** and **10** were both white amorphous powder with specific rotations of $[\alpha]_D^{20}$ +63.1 (*c* 0.10, MeOH) and +54.8 (*c* 0.10, MeOH), respectively. The HRESIMS (positive mode) quasi-molecular ion peaks at *m/z* 583.3235 ($[M+Na]^+$, Fig. S21A) and 581.3074 ($[M+Na]^+$,) give compounds **9** and **10** the molecular formulas of $C_{32}H_{48}O_8$ (Calculated for 583.3247) and $C_{32}H_{46}O_8$ (Calculated for 581.3090), respectively. The IR spectrum (Fig. S21B and Fig. S25B) showed strong absorption for hydroxy (3448 cm^{-1}), acyl carbonyl (1735 cm^{-1}) and carbonyl (1694 cm^{-1}) groups. A comparison of the NMR data of **9** (Fig. S22) and **10** (Tables 1 and 2, Fig. S26) showed high similarity of these two compounds. The only difference was that the carbon-carbon double bond ($\Delta^{23,24}$) was saturated in **9**, which led to the absence of a pair of olefinic carbon signals. An in-depth analysis into the NMR spectra of **10** demonstrated eight tertiary methyls [δ_H 0.87, 0.95, 1.12, 1.39, 1.42, 1.54, 1.56, 2.01 (3H each, s)]; three olefinic protons including a pair of *cis*-olefinic protons at δ_H 5.65 (1H, dd, J = 9.8, 3.0 Hz)/6.15 (1H, dd, J = 9.8, 1.5 Hz) and a pair of *trans*-olefinic protons at δ_H 6.43 (1H, d, J = 15.0 Hz)/7.03 (1H, d, J = 15.0 Hz); two carbonyl carbon signals at δ_C 202.6, 208.8; one carboxyl carbon signal at δ_C 170.4; two pairs of olefinic carbon signals at δ_C 133.3/129.0, 152.1/120.5; six oxygenated methines at δ_C 86.8, 79.4, 78.2, 75.3, 73.1, 71.8. These NMR data showed **10** highly resembled the known compound cucurbitacin C (Shang et al., 2014), which was also isolated in this study. Comparing with the 1D NMR data of cucurbitacin C, resonance signal diversities in **10** were observed. First, the chemical shift of C-5 moved to high-field at δ_C 86.8; secondly, the NMR signals attributed to the C-5/C-6 double bond was disappeared, instead, a pair of *cis*-olefinic protons at δ_H 6.15 (J = 9.8, 1.5 Hz)/5.65 (J = 9.8, 3.0 Hz) was observed; thirdly, the resonance signal of C-19 shifted Δ+14.1 ppm to the low-field at δ_C 75.2. These variations suggested the formation of an ether linkage between C-19 and C-5 position in **10**, which made the double bond between C-5 and C-6 transferred between C-6 and C-7. This deduction was further confirmed by the correlations from H-6, H-7 and H-19a (δ_H 3.38) to C-5 in the HMBC experiment (Fig. 3B and Fig. S27B). The stereochemistry of **10** was established by the biogenesis of cucurbitane backbone, as well as the NOESY experiments (Fig. S28) that key cross-peaks of H-19a (δ_H 3.38)/H-8 (δ_H 2.48), H-8/CH_3-18 (δ_H 1.12) were observed in **10**, which suggested 5,19-oxyrane ring and CH_3-18 of *β*-orientation. Thus, compound **10** was elucidated as 25-acetyloxy-3*β*,16*α*,20*β*-trihydroxy-5*β*, 19*β*-epoxy-cucurbit-6,23-dien-11,22-dione, and defined the name cucurbitacin C7. Likewise, compound **9**, the hydrogenated derivative of **10**, was elucidated as 25-acetyloxy-3*β*,16*α*,20*β*-trihydroxy-5*β*, 19*β*-epoxy-cucurbit-6-en-11,22-dione (cucurbitacin C6).

2.2. Bioactivities of the cucurbitacins

Although antitumor activities of cucurbitacins have been extensively studied (Blaskovich et al., 2003; Sun et al., 2005; Thoennissen et al., 2009), cytotoxic evaluation on cucurbitacin with a C-19 hydroxy group, especially the CuC types, is scarce (Sun et al., 2005; Wu et al., 2019). To enrich the potent reservoir of drug leads, MTT assays were employed to test the cytotoxicity of the cucurbitacins isolated from cucumber leaves. As described in Table 3, compounds **5**–**10** showed considerable growth inhibition capabilities against tumor cell lines HepG2, A549, DU145 and HCT116. Intriguingly, compounds **9** and **10** exhibited a significant inhibition effect compared to the positive control taxol (IC_{50} = 1.86 ± 0.17 μM) on HepG2 cell line with IC_{50} values of 10.06 ± 0.34 μM and 4.16 ± 0.42 μM, respectively. A preliminary analysis of the structure-activity relationships (SAR) of these cucurbitacins revealed that both the C-25-acetyl group and *α*,*β*-unsaturated ketone in the side chains were critical for their cytotoxicity. In contrast, the free hydroxy group on C-19 was deemed undesirable due to the stronger bioactivities of compounds **9** and **10** than those of compounds **7** and **8**. In addition, the apoptosis induction effect was observed when HepG2 cells were

Table 3
Antitumor activities of the cucurbitacins isolated from the leaves of cucumber.

Compounds	IC_{50}a (μM)			
	HepG2	A549	DU145	HCT116
1	n.d.	n.d.	70.12 ± 2.28	n.d.
2	n.d.	97.05 ± 3.18	42.19 ± 1.42	54.19 ± 1.81
3	62.17 ± 1.34	53.68 ± 2.36	48.52 ± 1.78	86.12 ± 1.22
4	85.62 ± 2.48	75.53 ± 1.96	36.37 ± 1.64	62.31 ± 1.93
5	40.15 ± 1.08	49.37 ± 1.33	25.44 ± 1.73	39.94 ± 1.07
6	37.26 ± 1.47	45.12 ± 1.07	18.26 ± 0.84	42.34 ± 0.94
7	16.42 ± 1.15	21.59 ± 0.89	12.54 ± 0.47	32.48 ± 1.16
8	9.52 ± 0.84	13.03 ± 0.40	8.12 ± 0.31	20.06 ± 0.34
9	10.06 ± 0.34	18.66 ± 0.87	3.61 ± 0.43	11.35 ± 0.52
10	4.16 ± 0.42	10.13 ± 0.78	1.82 ± 0.27	10.66 ± 0.37
taxol	1.84 ± 0.17	0.12 ± 0.01	0.16 ± 0.03b	6.29 ± 0.48b

[a]: IC_{50}, inhibition concentration at 50%. All values are presented as the mean ± SD (n = 3).[b]: nM. n.d.: IC_{50} > 100 μM.

exposed to four selected cucurbitacins (**7**–**10**) at two concentrations for 24 h, compounds **7**–**10** significantly induced HepG2 cells apoptosis at 10 μM (Fig. 4A and B).

Caspases are known to play a central role in apoptosis progress and are activated in a sequential cascade of cleavages from the inactive forms (Herr et al., 2001). Therefore, we examined the involvement of various caspases during cucurbitacins-induced apoptosis in HepG2 cells, and investigated the cleavages of caspase-8, caspase-9 and caspase-3 as well as the subsequent proteolytic cleavage of poly (ADP-ribose) polymerase (PARP) in HepG2 cells upon 10 μM cucurbitacins treatment at for 24 h. As shown in Fig. 4C and D, the levels of PARP, caspase-8, caspase-9 and caspase-3 were significantly decreased by the compounds **7**–**10** treatment, indicating that cucurbitacin-induced apoptosis in HepG2 cells is likely activated through a caspase-dependent pathway.

3. Conclusions

By phytochemical approaches, ten cucurbitacins including seven undescribed (compounds **1**–**5** and **9**–**10**) and three known ones (compounds **6**–**8**) have been isolated and structurally identified from the bitter leaves of cucumber. Their antitumor bioactivities were experimentally evaluated and discussed. Compounds **5**–**10** showed growth inhibition against tumor cell lines HepG2, A549, DU145 and HCT116, two cucurbitacins (**9** and **10**) exhibited a significant inhibition effect compared to the positive control taxol. The SAR analysis indicated that both the C-25-acetyl group and α,β-unsaturated ketone in the side chains were critical for their cytotoxicity. The likely antitumor mechanism of cucurbitacins was inducing tumor cell apoptosis and down-regulating the expression of caspase-related proteins. Phytochemical and bioactivity studies on cucurbitacin C type cucurbitacins would broaden their structural diversity and enrich the candidates for anticancer therapeutic

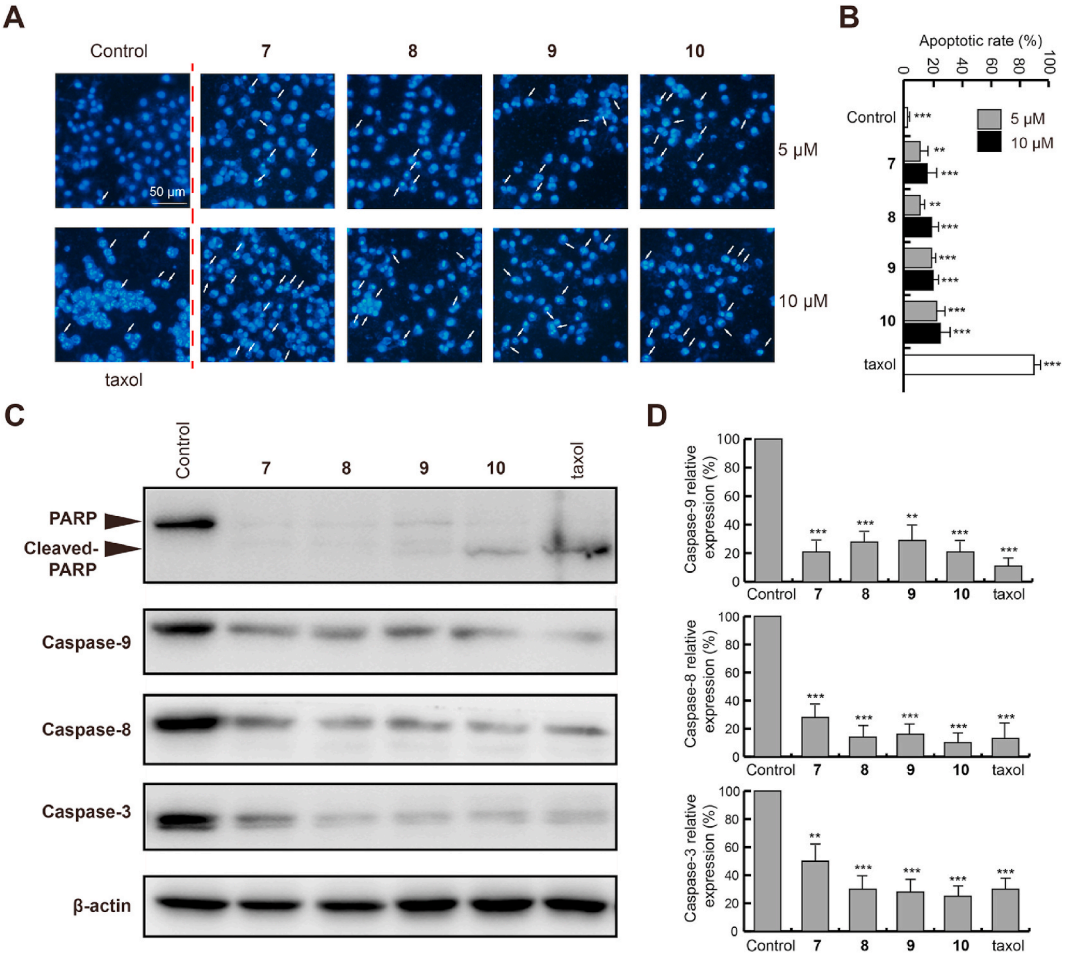


Fig. 4. Cucurbitacins induced caspase apoptosis pathway activation in HepG2 cells. (A) Apoptosis induction effects of four cucurbitacins on HepG2 cells. The cells were treated with each cucurbitacin for 24 h, and stained with Hoechst 33258 for cell apoptosis assay. Arrows, chromatin condensed HepG2 cells. (B) Quantitative data of panel A, percentage of apoptosis cells is shown. $^{**}P < 0.01$, $^{***}P < 0.001$, student t-test. (C) Inhibition effects of four cucurbitacins in caspase apoptosis pathway on HepG2 cells. The cells were treated with each cucurtibitcin for 24 h, respectively, and the major apoptosis-related proteins (PARP, cleaved PARP, Caspase-9, Caspase-8 and Caspase-3) were detected by Western blot. β-actin was used as internal reference. (D) Quantitative data of panel C, expression levels of apoptosis-related proteins are shown. $^{**}P < 0.01$, $^{***}P < 0.001$, student t-test.

agents.

4. Experimental

4.1. General experimental procedures

The HRESIMS for molecular formula determination was performed on an Agilent 1290 ultra-high performance liquid chromatography (UHPLC) combined with an Agilent 6540 quadrupole time-of-flight mass spectrometry (Q-TOF-MS) in positive ion mode. Optical rotations were measured on an Autopol-IV polarimeter (RudoPH Research analytical). IR spectra were obtained using a NEXUS-470 FTIR (Nicolet) spectrometer with KBr disks. 1D and 2D NMR experiments were carried out on a Bruker Avance III 500 system. Semi-preparative high-performance liquid chromatography (semi-HPLC) was performed on an Agilent 1260 infinity LC (Agilent ODS column, 250 × 9.4 mm i. d, 5 μm) equipped with a diode array detector (DAD). Thin-layer and column chromatography were performed using silica gel (Qingdao Haiyang Chemical Co. Ltd, GF_{254} or 200–300 mesh), Sephadex LH-20 (Pharmacia Biotech Ltd.), and C18 reversed-phase (RP-C18) (150–200 mesh, Merck, Germany). All the solvents of HPLC-grade were purchased from Fisher Scientific Company. Solvents for isolation were of analytical grade and were purchased from Beijing Chemical Co, Ltd.

4.2. Plant material

Fresh leaves of the *Cucumis sativus* L. (Cucurbitaceae) germ line "9930" were collected from the Vegetable Research Institute of Hunan Academy of Agricultural Sciences (GPS coordinates: 113°04′25.50″E, 28°10′46.12″N), Hunan province, China, in September 2020 and were authenticated by Prof. Huiming Cheng (Hunan Academy of Agricultural Sciences). A voucher specimen (HN20200924) has been deposited in the herbarium of Hunan Agricultural University.

4.3. Isolation of cucurbitacins from cucumber leaves

Fresh leaves (110 kg) of *C. sativus* were cut up and then extracted with 95% ethanol (3 × 600 L) at 90 °C in a 2000-liters extracting tank. After filtration, the extracted solutions were combined and evaporated under a vacuum to remove the ethanol. This extract (30 L) was defatted and decolorized using petroleum ether (PE, 60–90 °C; 8 × 20 L) and the final residue was subjected to column chromatography (CC) on the macroporous resin (D101, 2 kg, 60 × 14 cm) with an EtOH–H_2O gradient (20:80, 80:20, 95:5) to yield three fractions (Fr.1–Fr.3). Fr.2 (295 g) was chromatographed on silica gel (3.0 kg, 55 × 14 cm) with a CH_2Cl_2–MeOH gradient (20:1, 10:1, 5:1, 1:1) to yield seven fractions (Fr. A–Fr.G). Fr. A (12 g) was subjected to chromatography on silica gel (300 g, 40 × 5 cm) with an PE-EtOAc gradient (2:1, 1:1, 1:2) to afford five fractions (Fr.A-1–Fr.A-5). Fr. A-4 was chromatographed on RP-C18 silica gel (20 × 1.5 cm) in a gradient MeOH–H_2O elution system (50:50, 70:30, 100:0) to yield four fractions (Fr.A-4-1–Fr.A-4-4). Fr. A-4-2 was separated by semi-preparative HPLC (MeOH/H_2O, 57:43, 2.0 mL/min) to yield compounds **2** (4 mg, R_t = 21 min), **7** (520 mg, R_t = 28 min), **8** (7.1 g, R_t = 25 min), **9** (3 mg, R_t = 36 min), and **10** (4 mg, R_t = 34 min). Fr.B (5 g) was subjected to chromatography on silica gel (150 g, 30 × 4 cm) with a solvent of PE-EtOAc (1:2) to afford four fractions (Fr.B-1–Fr. B-4). Fr.B-2 was subjected to chromatography on RP-C18 silica gel (20 × 1.5 cm) with a gradient solvent of MeOH/H_2O (50:50, 70:30, 100:0) and then separated by semi-preparative HPLC (MeOH/H_2O, 55:45, 2.0 mL/min) to yield compounds **5** (62 mg, R_t = 23 min) and **6** (320 mg, R_t = 21 min). Fr.B-3 was chromatographed on Sephadex LH-20 (170 × 3 cm) in CH_2Cl_2–MeOH (1:1) and then separated by semi-preparative HPLC (MeOH/H_2O, 42:58, 2.0 mL/min) to yield compounds **1** (19 mg, R_t = 32 min) and **3** (5 mg, R_t = 37 min). Fr. D (10 g) was subjected to chromatography on silica gel (110 g, 20 × 4 cm) with a gradient solvent of CH_2Cl_2–MeOH (10:1, 5:1, 2:1) to yield three fractions (Fr.D-1–Fr.D-3). Fr. D-2 was then chromatographed on RP-C18 silica gel (30 × 2 cm) with a gradient solvent of MeOH/H_2O (50:50, 70:30, 100:0) to afford four fractions (Fr.D-2-1–Fr.D-2-4). Fr. D-2-2 was finally purified by semi-preparative HPLC (MeOH/H_2O, 38:62, 2.0 mL/min) to yield compound **4** (3 mg, R_T = 29 min).

Cucurbitacin C1 (1): white amorphous power; $[\alpha]_D^{20}$ +99.2 (*c* 0.10, MeOH); IR (KBr) ν_{max} 3416, 2970, 1697, 1644 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) and ^{13}C NMR (125 MHz, $CDCl_3$) data, see Tables 1 and 2; HRESIMS (positive) *m/z* 487.3409 $[M+H]^+$ (Calculated for $C_{30}H_{47}O_5$, 487.3418).

Cucurbitacin C2 (2): white amorphous power; $[\alpha]_D^{20}$ +70.0 (*c* 0.10, MeOH); IR (KBr) ν_{max} 3434, 2969, 1689 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) and ^{13}C NMR (125 MHz, $CDCl_3$) data, see Tables 1 and 2; HRESIMS (positive) *m/z* 537.3420 $[M+H]^+$ (Calculated for $C_{30}H_{49}O_8$, 537.3427).

Cucurbitacin C3 (3): white amorphous power; $[\alpha]_D^{20}$ +54.7 (*c* 0.10, MeOH); IR (KBr) ν_{max} 3393, 2976, 1726, 1700 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) and ^{13}C NMR (125 MHz, $CDCl_3$) data, see Tables 1 and 2; HRESIMS (positive) *m/z* 607.3835 $[M+H]^+$ (Calculated for $C_{34}H_{55}O_9$, 607.3841).

Cucurbitacin C4 (4): white amorphous power; $[\alpha]_D^{20}$ +112.0 (*c* 0.10, MeOH); IR (KBr) ν_{max} 3435, 2971, 1694 cm^{-1}; ^1H NMR (500 MHz, C_5D_5N) and ^{13}C NMR (125 MHz, C_5D_5N) data, see Tables 1 and 2; HRESIMS (positive) *m/z* 551.3574 $[M+H]^+$ (Calculated for $C_{31}H_{51}O_8$, 551.3578).

Cucurbitacin C5 (5): white amorphous power; $[\alpha]_D^{20}$ +85.9 (*c* 0.10, MeOH); IR (KBr) ν_{max} 3400, 2969, 1697, 1643 cm^{-1}; ^1H NMR (500 MHz, C_5D_5N) and ^{13}C NMR (125 MHz, C_5D_5N) data, see Tables 1 and 2; HRESIMS (positive) *m/z* 521.3471 $[M+H]^+$ (Calculated for $C_{30}H_{49}O_7$, 521.3473).

Cucurbitacin C6 (6): white amorphous power; $[\alpha]_D^{20}$ +63.1 (*c* 0.10, MeOH); IR (KBr) ν_{max} 3448, 2974, 2943, 1697 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) and ^{13}C NMR (125 MHz, $CDCl_3$) data, see Tables 1 and 2; HRESIMS (positive) *m/z* 583.3235 $[M+Na]^+$ (Calculated for $C_{32}H_{48}O_8Na$, 583.3247).

Cucurbitacin C7 (7): white amorphous power; $[\alpha]_D^{20}$ +54.8 (*c* 0.10, MeOH); IR (KBr) ν_{max} 3447, 2975, 2943, 1735, 1694, 1630 cm^{-1}; ^1H NMR (500 MHz, $CDCl_3$) and ^{13}C NMR (125 MHz, $CDCl_3$) data, see Tables 1 and 2; HRESIMS (positive) *m/z* 581.3074 $[M+Na]^+$ (Calculated for $C_{32}H_{46}O_8Na$, 581.3090).

4.4. Computational methods

To compute the vicinal proton-proton dihedral angle and proton chemical shift, MD simulations on enantiomers 24*R*-**2** and 24*S*-**2** were implemented at the GFN1-xTB level in vacuum using ORCA (Neese, 2017) to generate a total of 2000 snapshots in 1ns MD simulation. Then, the snapshots were geometry optimized at the GFN2-xTB level in vacuum using ORCA, and conformers with an energy cutoff of 5 kcal/mol were chosen for a subsequent DFT treatment. Subsequently, geometry optimizations and frequency analyses were implemented at the B3LYP-D3(BJ)/6-31G(d) level in PCM chloroform by the Gaussian09 program (Frisch et al., 2009). Gibbs free energies for conformers were determined by using thermal correction at the B3LYP-D3(BJ)/6-31G (d) level in PCM chloroform using Gaussian09 program and electronic energies evaluated at the wB97M-V/def2-TZVP level in PCM chloroform using ORCA. Boltzmann weights were computed using relative Gibbs free energies. For dihedral angle analysis, the vicinal proton-proton dihedral angles were measured using VMD. For proton chemical shift analysis, NMR shielding constants were computed using the GIAO method at the B972/pcSseg-2 level in PCM chloroform by the Gaussian09 program.

4.5. MTT assay

The cell lines of HepG2, A549, Du145, HCT116 were purchased from Peking Union Medical College, Cell Bank (Beijing, China). These cells were maintained in DMEM medium (Hyclone, Waltham, MA, USA) supplemented with 10% fetal bovine serum (Hyclone), penicillin (100 U/mL) and streptomycin (100 μg/mL) at 37 °C in a humidified incubator with 95% air and 5% CO_2. The cells were treated with the isolated compounds or positive drug taxol (Sigma-Aldrich, USA) for 24 h. The medium containing 0.5 mg/mL 3-(4,5-dimethylthiazolyl-2)-2,5-diphenyltetrazolium bromide (MTT) was added, and the cells were incubated for 4 h at 37 °C. Subsequently, the medium was removed, and 100 μL DMSO was added. The absorbance at 550 nm was detected using a microplate reader. Cell viability was expressed as the mean percentage of absorbance in treated vs. control cells. Inhibition rate = 1-[$OD_{(treatment)}$-$OD_{(background)}$]/[$OD_{(control)}$/$OD_{(background)}$] × 100%.

4.6. Western blotting

HepG2 cells were treated with different compounds (10 μM) or taxol (0.1 μM) for 24 h. Then cells were washed with PBS three times and lysed in RIPA lysis buffer containing a cocktail of protease inhibitors (Roche). Total proteins were isolated, separated by SDS-PAGE, and transferred to polyvinylidene fluoride (PVDF) membranes. After that, the membranes were blocked with 5% non-fat milk and incubated with primary antibodies for 2 h at 4 °C, followed by incubating with anti-rabbit IgG conjugated to HRP (1:1000 dilution) at room temperature for 1 h. The membranes were developed with Immobilon Western Chemiluminescent HRP Substrate, and scanned with Tanon 5200 Imaging Analysis System (Tanon, Shanghai, China).

4.7. Hoechst 33258 assay

HepG2 cells were treated with different compounds (5 and 10 μM) for 24 h and then fixed in 4% paraformaldehyde for 30 min at room temperature. For Hoechst 33258 staining, HepG2 cells were incubated with 500 μL of Hoechst 33258 (10 μg/mL) for 10 min at 37 °C. After three washes with PBS, the cells were sealed on slides. Fluorescence images were obtained using a fluorescence microscope (Olympus, Tokyo, Japan) at 350 nm (excitation) and 461 nm (emission) wavelengths.

4.8. Statistical analysis

All of the results are expressed as mean ± standard deviation. Statistical analysis was performed with Student's t-test for comparison of two groups. In both cases, differences with $P < 0.05$ were statistically significant.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.phytochem.2022.113123.

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